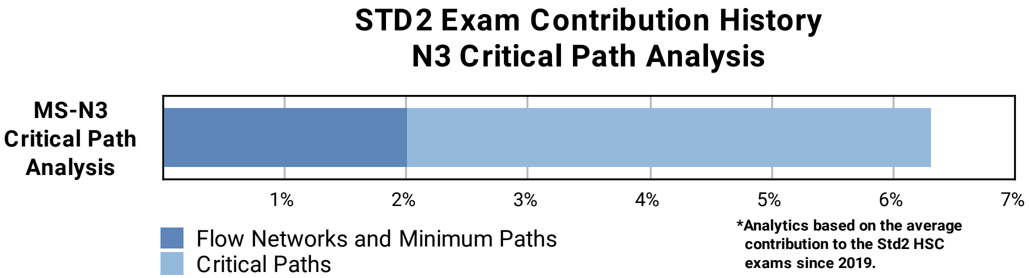
STD 2: Networks (Std 2), N3 Critical Path Analysis (Y12)

Critical Paths

Teacher: Davidson Lai

Exam Equivalent Time: 178.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*



OVERVIEW

- *Networks* is a new Std2 topic area and has been allocated between 9-13% of the Std2 exam over the first 3 years of the new syllabus.
- We have split *N3 Critical Path Analysis (6.3%)* into two categories for the purposes of analysis: *1-Critical Paths (4.3%) and 2-Flow Networks and Minimum Cuts (2.0%).*
- This analysis will look at *Critical Paths,* the largest and most challenging sub-topic within *Networks.*

ANALYSIS

- *Critical Paths* warrants a significant revision focus, with mark allocations of 4, 5 and 4 in the 2021-2019 Std2 exams respectively. The core skills students need in this area are set out below:
- Skill 1: Draw a network diagram from a table. Tested with 6 activities in the 2019 HSC exam, and well answered. Numerous questions in the database expose students to many of the variations in this area.
- Skill 2: Scanning forwards (only) to establish critical paths.
- Skill 3: An extension of skill 2 that requires the scanning forwards *and* backwards of a network diagram. This exercise is the most challenging and provides the EST and LST data to calculate float times and interrogate network adjustments.
- Float time questions have caused major problems in each year of the new syllabus, producing sub-35% mean marks - a key revision focus area.
- The importance of *setting out* "scanning" information in network diagrams logically and clearly cannot be overstated, and detailed worked solutions are provided to underscore this. Note that two styles of answer layout are provided - one reflecting the NESA sample exam solution and the other following the *Networks Topic Guidance*.
- Questions involving shortening critical paths have proven challenging, particularly when reducing float times create new critical path(s). A number of questions explore this concept.

Questions

1. Networks, STD2 N3 2006 FUR1 5 MC

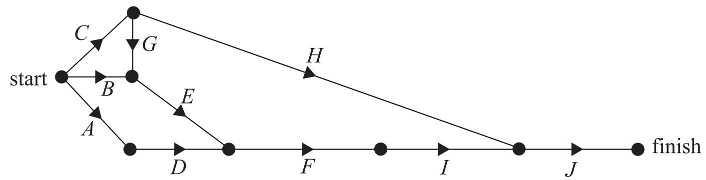
For a particular project there are ten activities that must be completed.

These activities and their immediate predecessors are given in the following table.

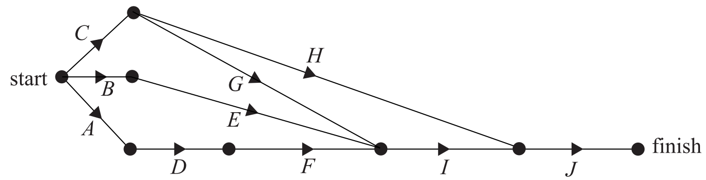
Activity	Immediate predecessors
A	–
B	–
C	–
D	A
E	B
F	D, E
G	C
H	C
I	F, G
J	H, I

A directed graph that could represent this project is

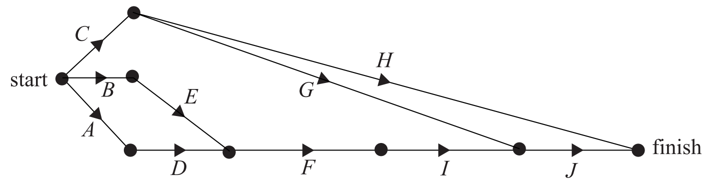
A.



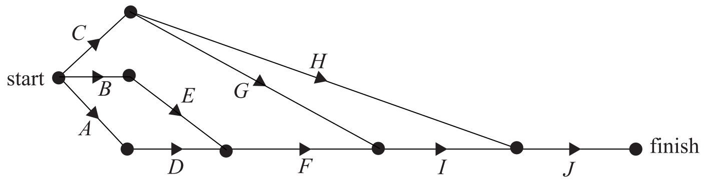
B.



C.

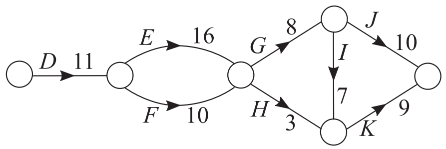


D.



2. Networks, STD2 N3 SM-Bank 38 MC

Identify the critical path through this network.



A. $DEGJ$

B. $DFHK$

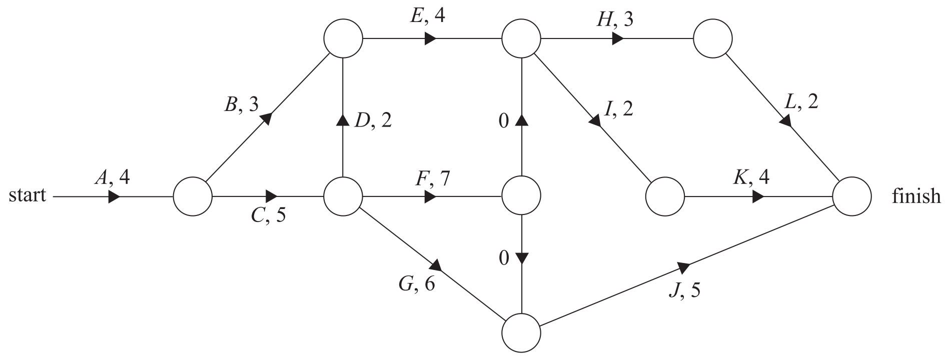
C. $DFGIK$

D. $DEGIK$

3. Networks, STD2 N3 2011 FUR1 8 MC

The diagram shows the tasks that must be completed in a project.

Also shown are the completion times, in minutes, for each task.



The critical path for this project includes activities

A. B and I.

B. C and H.

C. D and E.

D. F and K.

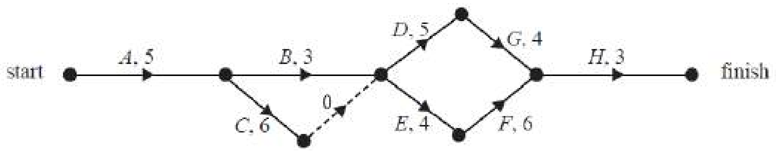
4. Networks, STD2 N3 2015 FUR1 9 MC

The table below shows, in minutes, the duration, the earliest starting time (EST) and the latest starting time (LST) of eight activities needed to complete a project.

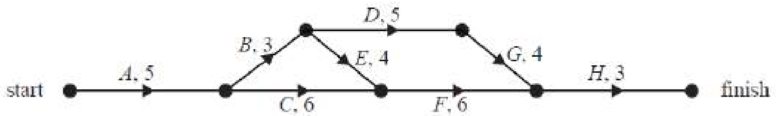
Activity	Duration	EST	LST
A	5	0	0
B	3	5	5
C	6	5	6
D	5	8	9
E	4	8	8
F	6	12	12
G	4	13	14
H	3	18	18

Which one of the following directed graphs shows the sequence of these activities?

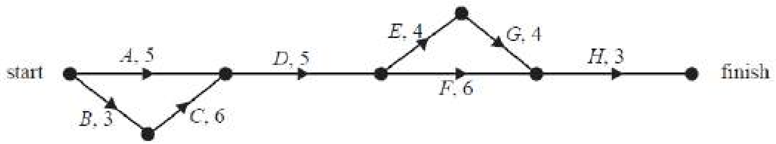
A.



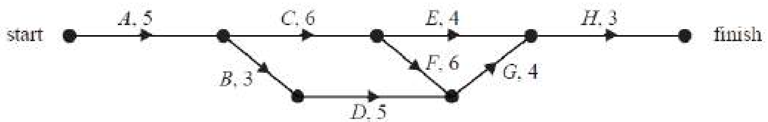
B.



C.

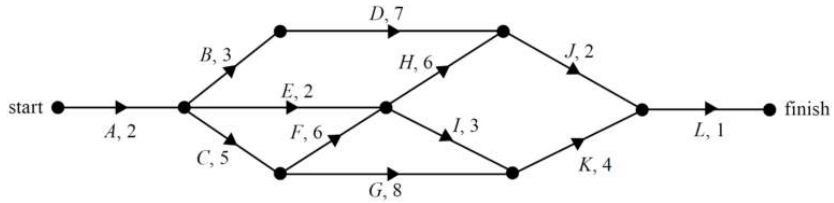


D.



5. Networks, STD2 N3 2008 FUR1 8-9 MC

The network below shows the activities that are needed to finish a particular project and their completion times (in days).



Part 1

The earliest start time for Activity K, in days, is

A. 7

B. 15

C. 16

D. 19

Part 2

This project currently has one critical path.

A second critical path, in addition to the first, would be created by

A. increasing the completion time of D by 7 days.

B. increasing the completion time of G by 1 day.

C. increasing the completion time of I by 2 days.

D. decreasing the completion time of C by 1 day.

6. Networks, STD2 N3 2011 FUR1 7 MC

Andy, Brian and Caleb must complete three activities in total (*K*, *L* and *M*)

The table shows the person selected to complete each activity, the time it will take to complete the activity in minutes and the immediate predecessor for each activity.

Person	Activity	Duration	Immediate predecessor
Andy	*K*	13	–
Brian	*L*	5	*K*
Caleb	*M*	16	*L*

All three activities must be completed in a total of 40 minutes.

The instant that Andy starts his activity, Caleb gets a telephone call.

The maximum time, in minutes, that Caleb can speak on the telephone before he must start his allocated activity is

A. 5

B. 13

C. 18

D. 24

7. Networks, STD2 N3 2014 FUR1 8 MC

Which one of the following statements about critical paths is **true**?

A. There can be only one critical path in a project.

B. A critical path will always include the activity that takes the longest time to complete.

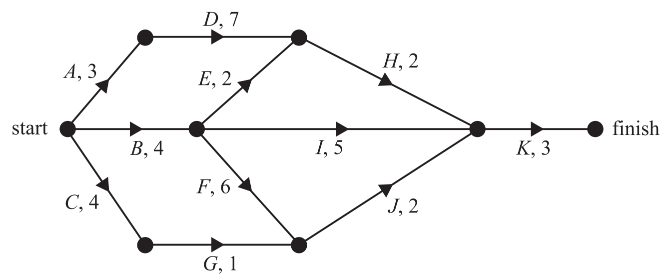
C. Reducing the time of any activity on a critical path for a project will always reduce the minimum completion time for the project.

D. If there are no other changes, increasing the time of any activity on a critical path will always increase the completion time of a project.

8. Networks, STD2 N3 2018 FUR1 5 MC

The directed network below shows the sequence of 11 activities that are needed to complete a project.

The time, in weeks, that it takes to complete each activity is also shown.



How many of these activities could be delayed without affecting the minimum completion time of the project?

A. 3

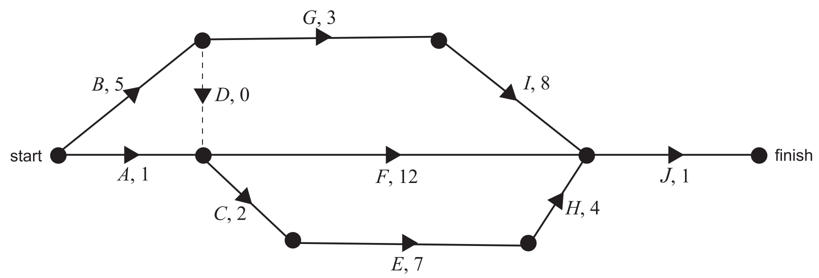
B. 4

C. 5

D. 6

9. Networks, STD2 N3 2006 FUR1 9 MC

The network below shows the activities and their completion times (in hours) that are needed to complete a project.



The project is to be optimised by reducing the completion time of **one** activity only.

This will reduce the completion time of the project by a maximum of

A. 1 hour

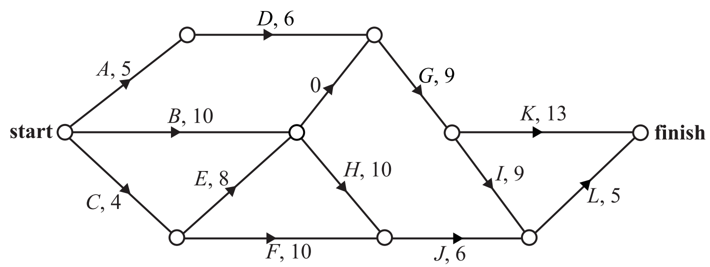
B. 3 hours

C. 4 hours

D. 5 hours

10. Networks, STD2 N3 2010 FUR1 8 MC

A project has 12 activities. The network below gives the time (in hours) that it takes to complete each activity.



The critical path for this project is

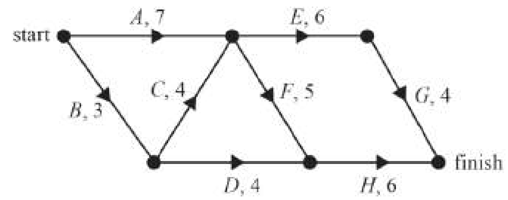
A. $ADGK$

B. $ADGIL$

C. $BHJL$

D. $CEGIL$

11. Networks, STD2 N3 2012 FUR1 8 MC



Eight activities, A, B, C, D, E, F, G and H, must be completed for a project.

The network above shows these activities and their usual duration in hours.

The duration of each activity can be reduced by one hour.

To complete this project in 16 hours, the minimum number of activities that must be reduced by one hour each is

A. 1

B. 2

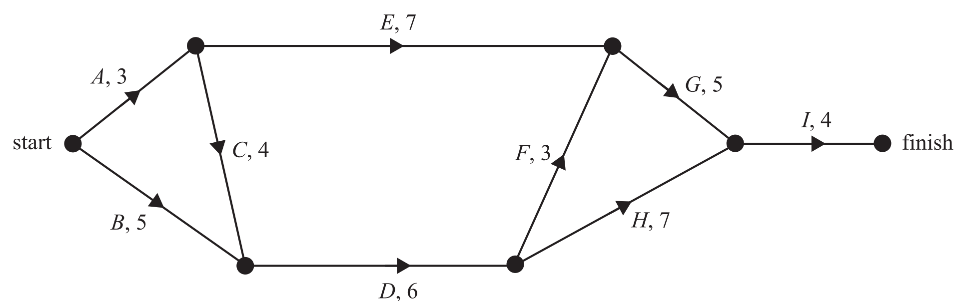
C. 3

D. 4

12. Networks, STD2 N3 2019 FUR2-N 2

The construction of the new reptile exhibit is a project involving nine activities, A to I.

The directed network below shows these activities and their completion times in weeks.



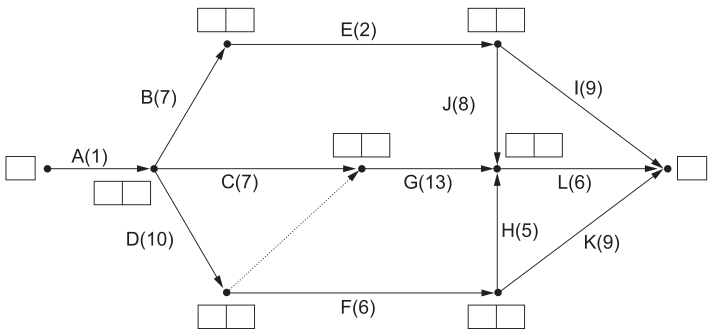
a. Which activities have more than one immediate predecessor? ***(1 mark)***

b. Write down the critical path for this project. ***(1 mark)***

c. What is the latest start time, in weeks, for activity B? ***(1 mark)***

13. Networks, STD2 N3 SM-Bank 31

Murray is building a new garage. The project involves activities A to L.



The network diagram shows these activities and their completion times in days.

a. Which TWO activities immediately precede activity G? *(1 mark)*

b. By completing the diagram shown, calculate the minimum time required to build the new garage. *(2 marks)*

c. Hence, what is the float time for activity E? *(1 mark)*

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14. Networks, STD2 N3 SM-Bank 33

A project requires nine activities (*A*–*I*) to be completed. The duration, in hours, and the immediate predecessor(s) of each activity are shown in the table below.

Activity	Duration (hours)	Immediate predecessor(s)
A	4	–
B	3	*A*
C	7	*A*
D	2	*A*
E	5	*B*
F	2	*C*
G	4	*E, F*
H	5	*D*
I	3	*G, H*

i. Sketch the network, identifying each activity and its duration. *(2 marks)*

ii. Identify the critical path and the minimum completion time of the project. *(2 marks)*

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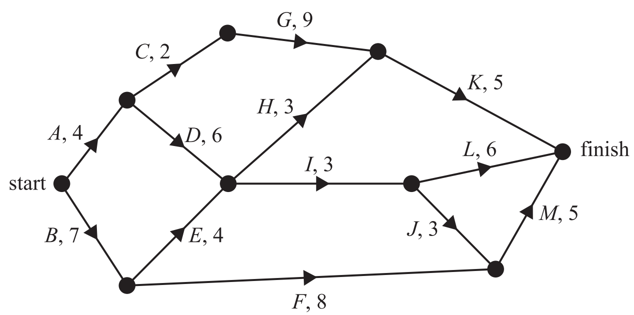
15. Networks, STD2 N3 FUR1 2020 6

The activity network below shows the sequence of activities required to complete a project.

The number next to each activity in the network is the time it takes to complete that activity, in days.



What is the critical path and minimum completion time for this project. *(2 marks)*

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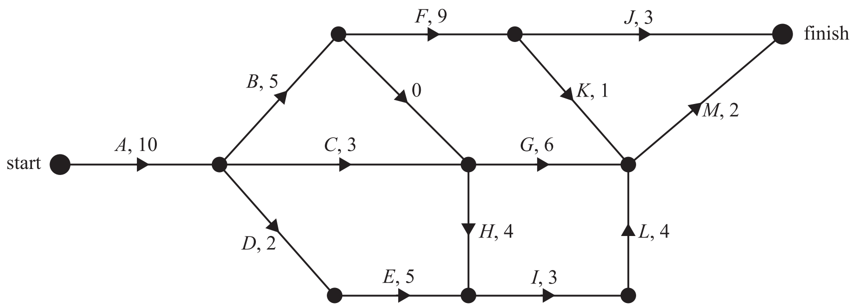
16. Networks, STD2 N3 2012 FUR2 2

Thirteen activities must be completed before the produce grown on a farm can be harvested.

The directed network below shows these activities and their completion times in days.



a. Determine the earliest starting time, in days, for activity E. *(1 mark)*

b. An activity with zero duration starts at the end of activity B.
Explain why this activity is used on the network diagram. *(1 mark)*

c. Determine the earliest starting time, in days, for activity H. *(1 mark)*

d. In order, list the activities on the critical path. *(1 mark)*

e. Determine the latest starting time, in days, for activity J. *(1 mark)*

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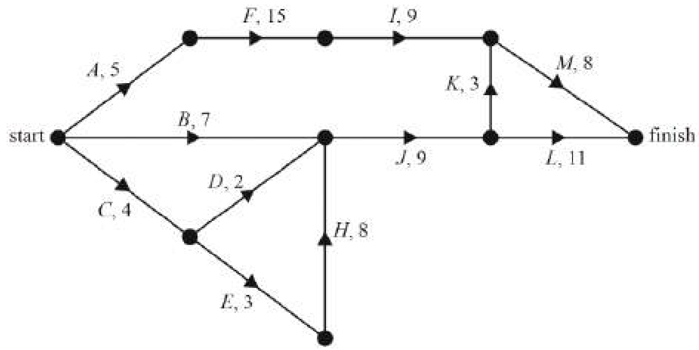
17. Networks, STD2 N3 2013 FUR2 2

A project will be undertaken in the wildlife park. This project involves the 13 activities shown in the table below. The duration, in hours, and predecessor(s) of each activity are also included in the table.

Activity	Duration (hours)	Predecessor(s)
A	5	–
B	7	–
C	4	–
D	2	C
E	3	C
F	15	A
G	4	B, D, H
H	8	E
I	9	F, G
J	9	B, D, H
K	3	J
L	11	J
M	8	I, K

Activity G is missing from the network diagram for this project, which is shown below.



a. Complete the network diagram above by inserting activity G. *(1 mark)*

b. Determine the earliest starting time of activity H. *(1 mark)*

c. Given that activity G is not on the critical path

i. write down the activities that are on the critical path in the order that they are completed *(1 mark)*

ii. find the latest starting time for activity D. ***(1 mark)***

d. Consider the following statement.

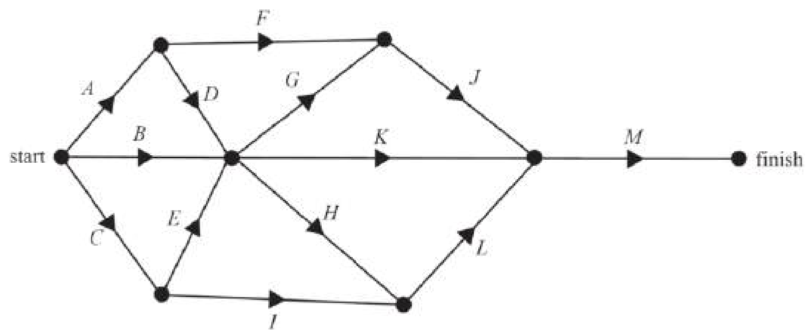
'If the time to complete just one of the activities in this project is reduced by one hour, then the minimum time to complete the entire project will be reduced by one hour.'

Explain the circumstances under which this statement will be true for this project. ***(1 mark)***

e. Assume activity F is reduced by two hours.
What will be the minimum completion time for the project? ***(1 mark)***

18. Networks, STD2 N3 SM-Bank 48

The network shows the activities that are needed to complete a particular project.



i. Find the total number of activities that need to be completed before activity L can begin. ***(1 mark)***

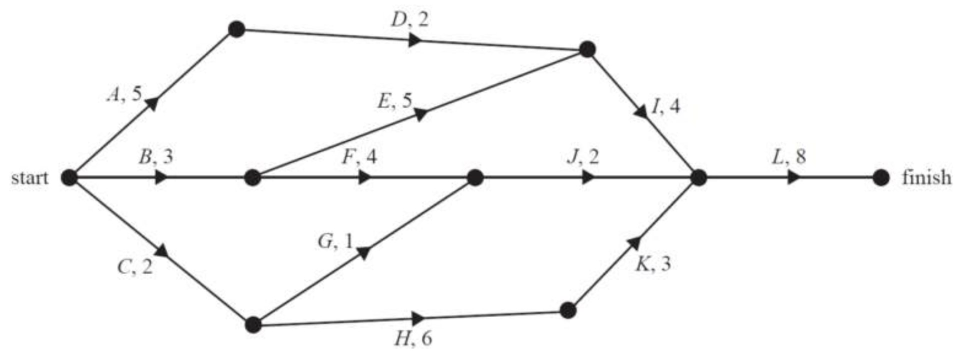
The duration of every activity is initially 5 hours.

ii. If the completion times of both activity F and activity K are reduced to 3 hours each, calculate the effect on the completion time for the project. ***(2 marks)***

19. Networks, STD2 N3 SM-Bank 47

The directed graph below shows the sequence of activities required to complete a project.

All times are in hours.



i. Find the number of activities that have exactly two immediate predecessors. ***(1 mark)***

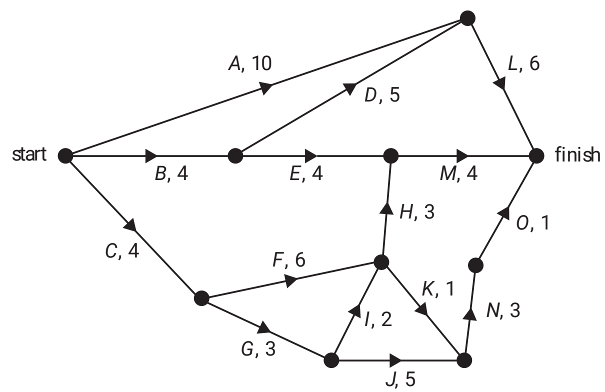
ii. Identify the critical path for this project. ***(3 marks)***

iii. If Activity E is reduced by one hour, identify the two new critical paths. ***(1 mark)***

20. Networks, STD2 N3 SM-Bank 49

The directed graph below shows the sequence of activities required to complete a project.

The time to complete each activity, in hours, is also shown.



i. Find the earliest starting time, in hours, for activity N. ***(2 marks)***

To complete the project in minimum time, some activities cannot be delayed.

ii. Calculate the number of activities that cannot be delayed. ***(1 mark)***

21. Networks, STD2 N3 2006 FUR2 3

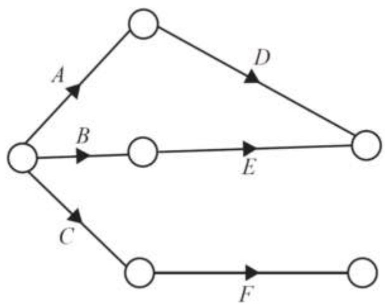
The five musicians are to record an album. This will involve nine activities.

The activities and their immediate predecessors are shown in the following table.

The duration of each activity is not yet known.

Activity	Immediate predecessors
A	–
B	–
C	–
D	A
E	B
F	C
G	D, E
H	F
I	G, H

a. Use the information in the table above to complete the network below by including activities G, H and I. *(2 marks)*



There is only one critical path for this project.

b. How many **non-critical** activities are there? *(1 mark)*

The following table gives the earliest start times (EST) and latest start times (LST) for **three of the activities only.** All times are in hours.

Activity	EST	LST
A	0	2
C	0	1
I	12	12

c. Write down the critical path for this project. *(1 mark)*

The minimum time required for this project to be completed is 19 hours.

d. What is the duration of activity I? *(1 mark)*

The duration of activity C is 3 hours.

e. Determine the maximum combined duration of activities F and H. *(1 mark)*

22. Networks, STD2 N3 2019 HSC 26

A project requires activities A to F to be completed. The activity chart shows the immediate prerequisite(s) and duration for each activity.

Activity	*Immediate prerequisite(s)*	*Duration in hours*
A	—	2
B	A	6
C	A	5
D	B	2
E	C, D	4
F	E	1

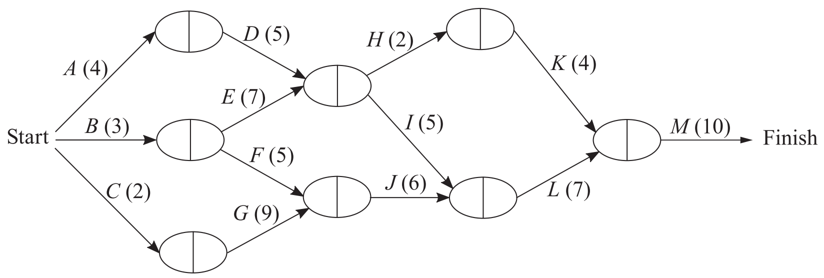
a. By drawing a network diagram, determine the minimum time for the project to be completed. *(3 marks)*

b. Determine the float time of the non-critical activity. *(1 mark)*

23. Networks, STD2 N3 SM-Bank 39

Bianca is designing a project for producing an advertising brochure. It involves activities *A-M*.

The network below shows these activities and their completion time in hours.



i. What is the earliest starting time of activity *J*? ***(1 mark)***

ii. What is the latest starting time of activity *H*? ***(1 mark)***

iii. What is the float time of activity *I*? ***(1 mark)***

iv. What is the minimum time required to produce the brochure? ***(2 marks)***

24. Networks, STD2 N3 SM-Bank 49

A project requires nine activities (*A–I*) to be completed. The duration, in hours, and the immediate predecessor(s) of each activity are shown in the table below.

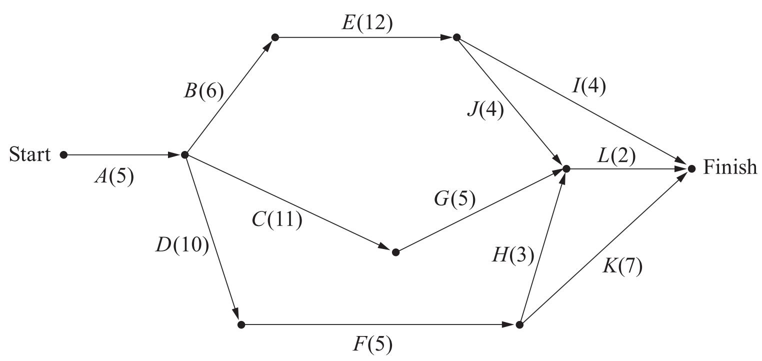
Activity	Duration (hours)	Immediate predecessor(s)
A	4	–
B	3	*A*
C	7	*A*
D	2	*A*
E	5	*B*
F	2	*C*
G	4	*E*, *F*
H	5	*D*
I	3	*G*, *H*

i. Sketch the network diagram. ***(3 marks)***

ii. Find the minimum completion time for this project, in hours. ***(1 mark)***

25. Networks, STD2 N3 SM-Bank 29

The construction of The Royal Easter show involves activities A to L. The diagram shows these activities and their completion times in days.

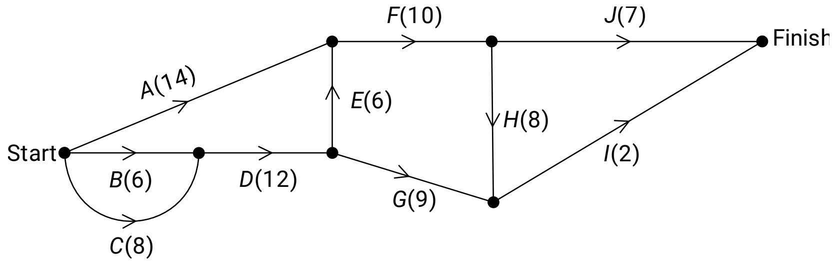


The company contracted to construct it are given a completion deadline of 31 days.

Calculate the float time of Activity G. ***(3 marks)***

26. Networks, STD2 N3 2020 HSC 26

The preparation of a meal requires the completion of all activities A to J. The network diagram shows the activities and their completion times in minutes.



a. What is the minimum time needed to prepare the meal? ***(1 mark)***

b. List the activities which make up the critical path for this network. ***(2 marks)***

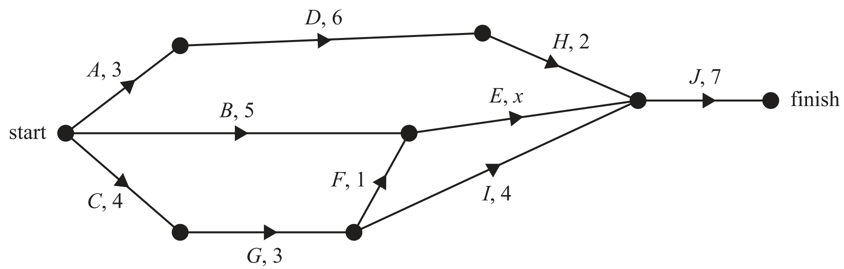
c. Complete the table below, showing the earliest start time and float time for activities A and G ***(2 marks)***

Activity	*Earliest start time* (minutes)	*Float time* (minutes)
A		
G		

27. Networks, STD2 N3 2021 FUR1 6

The directed graph below shows the sequence of activities required to complete a project.

The time taken to complete each activity, in hours, is also shown.



The minimum completion time for this project is 18 hours.

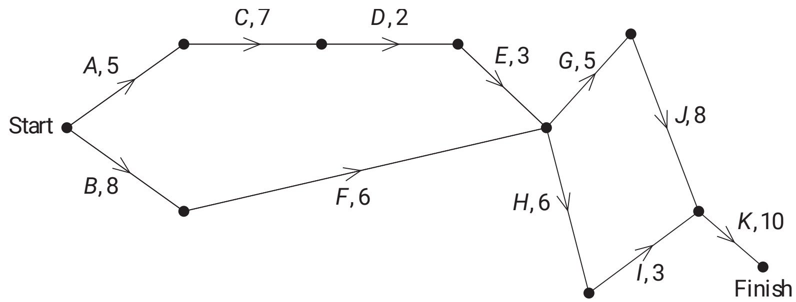
The time taken to complete activity E is labelled x.

What is the maximum value of x? ***(2 marks)***

28. Networks, STD2 N3 2021 HSC 36

A project requires completion of 11 tasks $A, B, C, \ldots, K.$

A network diagram for the project giving the completion time for each task, in minutes, is shown.



a. Find the minimum time to complete the project. ***(1 mark)***

b. State the critical path for this project. ***(1 mark)***

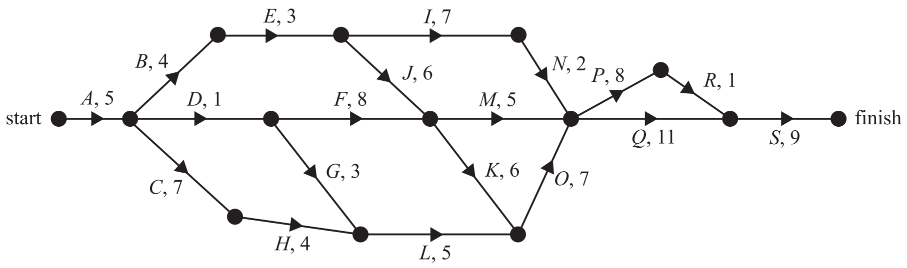
c. A new task, X, is to be added to the project. The earliest starting time for X is 17 minutes, the latest starting time for X is 18 minutes and X has a completion time of 12 minutes.

Add task X to the given network diagram above AND state the float time for this task. ***(3 marks)***

29. Networks, STD2 N3 SM-Bank 41

The directed network below shows the sequence of activities, A to S, that is required to complete a manufacturing process.

The time taken to complete each activity, in hours, is also shown.



i. Determine the critical path of this network. ***(2 marks)***

ii. Identify the activities that have a float time of 10 hours. ***(2 marks)***

30. Networks, STD2 N3 SM-Bank 44

A project involves nine activities, A to I.

The immediate predecessor(s) of each activity is shown in the table below.

Activity	Immediate predecessor(s)
A	–
B	A
C	A
D	B
E	B, C
F	D
G	D
H	E, F
I	G, H

A directed network for this project will require a dummy activity.

Sketch the network diagram, clearly identifying the dummy activity. ***(3 marks)***

31. Networks, STD2 N3 EQ-Bank 18

An engineering project requires activities *A* to *G* to be completed, as shown in the table.

Activity	*Immediate prerequisites(s)*	*Duration in weeks*
A	-	10
B	-	?
C	*A, B*	15
D	*B*	?
E	*C*	12
F	*C*	?
G	*D, E*	?

The minimum completion time for the project is 40 weeks and the critical path includes activities *A*, *C*, *E* and *G*. The float for activity *F* is six weeks and the float for activity *D* is 9 weeks.

Find the possible duration for each of the activities *B*, *D*, *F* and *G*. Include a network diagram in your answer. ***(5 marks)***

32. Networks, STD2 N3 EQ-Bank 19

An engineering project requires activities *A* to *G* to be completed, as shown in the table.

Activity	*Immediate prerequisites(s)*	*Duration in days*
A	-	?
B	-	7
C	*A*	?
D	*A, B*	12
E	*D*	5
F	*C, E*	?
G	*D*	?

The minimum completion time for the project is 25 days and the critical path includes activities *B*, *D*, *E* and *F*. The float for activity *G* is two days and the float for activity *C* is four days.

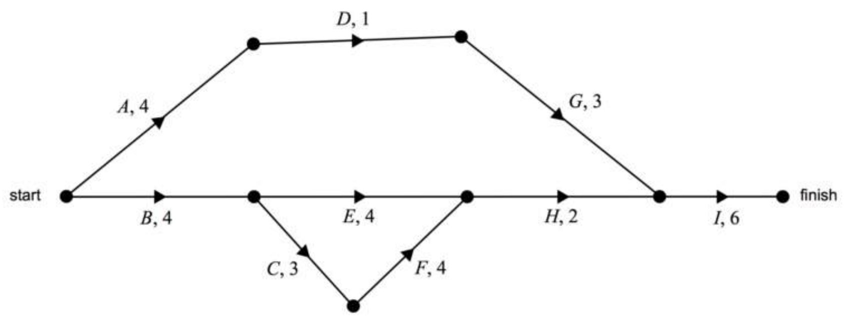
Find the possible duration for each of the activities *A*, *C*, *F* and *G*. Include a network diagram in your answer. ***(5 marks)***

33. Networks, STD2 N3 2007 FUR2 4

A community centre is to be built on the new housing estate.

Nine activities have been identified for this building project.

The directed network below shows the activities and their completion times in weeks.



a. Determine the minimum time, in weeks, to complete this project. ***(1 mark)***

b. Determine the float time, in weeks, for activity D. ***(2 marks)***

The builders of the community centre are able to speed up the project.

Some of the activities can be reduced in time at an additional cost.

The activities that can be reduced in time are A, C, E, F and G.

c. Which of these activities, if reduced in time individually, would **not** result in an earlier completion of the project? ***(1 mark)***

The owner of the estate is prepared to pay the additional cost to achieve early completion.

The cost of reducing the time of each activity is $5000 per week.

The maximum reduction in time for each one of the five activities, A, C, E, F, G, is 2 weeks.

d. Determine the minimum time, in weeks, for the project to be completed now that certain activities can be reduced in time. ***(1 mark)***

e. Determine the minimum additional cost of completing the project in this reduced time. ***(1 mark)***

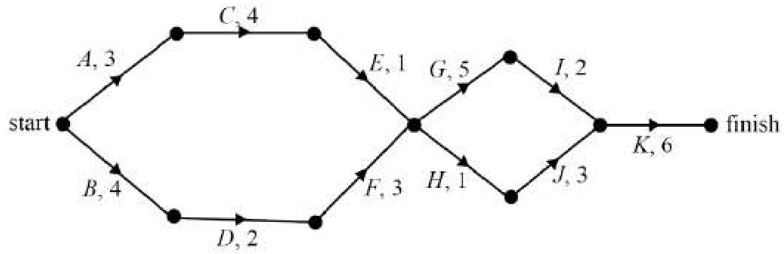
---

34. Networks, STD2 N3 2009 FUR2 4

A walkway is to be built across the lake.

Eleven activities must be completed for this building project.

The directed network below shows the activities and their completion times in weeks.



a. What is the earliest start time for activity *E*? ***(1 mark)***

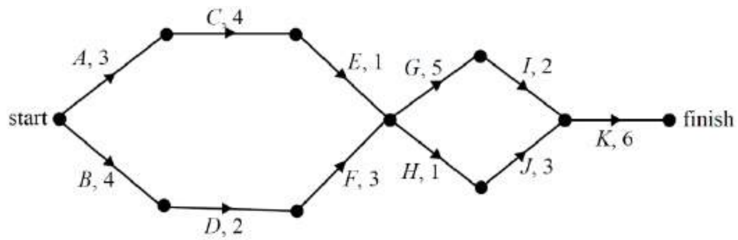
b. Write down the critical path for this project. ***(1 mark)***

c. The project supervisor correctly writes down the float time for each activity that can be delayed and makes a list of these times.

Determine the longest float time, in weeks, on the supervisor's list. ***(1 mark)***

A twelfth activity, *L*, with duration three weeks, is to be added without altering the critical path.

Activity *L* has an **earliest** start time of four weeks and a **latest** start time of five weeks.



d. Draw in activity *L* on the network diagram above. ***(1 mark)***

e. Activity *L* starts, but then takes four weeks longer than originally planned.

Determine the total overall time, in weeks, for the completion of this building project. ***(1 mark)***

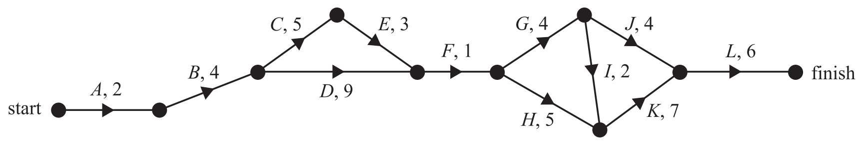
---

35. Networks, STD2 N3 2019 FUR2 3

Fencedale High School is planning to renovate its gymnasium.

This project involves 12 activities, A to L.

The directed network below shows these activities and their completion times, in weeks.



The minimum completion time for the project is 35 weeks.

a. Identify the critical path and state how many activities are on it? ***(2 marks)***

b. Determine the latest start time of activity E. ***(1 mark)***

c. Which activity has the longest float time? ***(1 mark)***

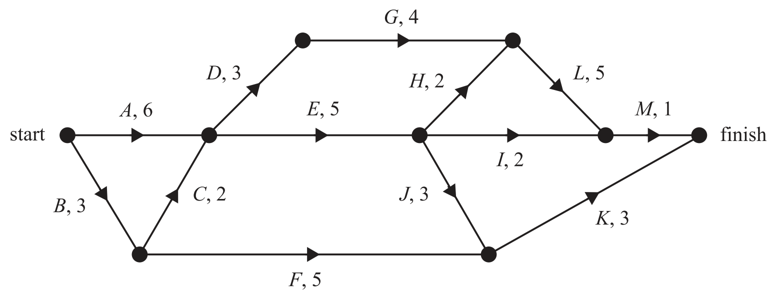
It is possible to reduce the completion time for activities C, D, G, H and K by employing more workers.

d. The completion time for each of these five activities can be reduced by a maximum of two weeks.
What is the minimum time, in weeks, that the renovation project could take? ***(1 mark)***

36. Networks, STD2 S3 SM-Bank 50

Roadworks planned by the local council require 13 activities to be completed.

The network below shows these 13 activities and their completion times in weeks.



a. What is the earliest start time, in weeks, of activity K? ***(1 mark)***

b. How many of these activities have zero float time? ***(2 marks)***

c. It is possible to reduce the completion time for activities A, E, F, L and K.

The reduction in completion time for each of these five activities will incur an additional cost.

The table below shows the five activities that can have their completion time reduced and the associated weekly cost, in dollars.

Activity	Weekly cost ($)
A	140 000
E	100 000
F	100 000
L	120 000
K	80 000

The completion time for each these five activities can be reduced by a maximum of two weeks.

The overall completion time for the roadworks can be reduced to 16 weeks.

What is the minimum cost, in dollars, of this change in completion time? ***(3 marks)***

Worked Solutions

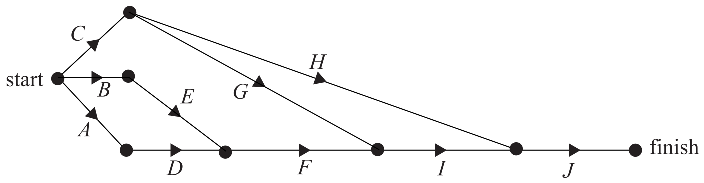
1. Networks, STD2 N3 2006 FUR1 5 MC

Consider the information in the table:

D and E are prerequisites to F $\rightarrow$ eliminate B

G is a prerequisite to I $\rightarrow$ eliminate A and C

$\Rightarrow$ Option D reflects the information in the table:



$\Rightarrow D$

2. Networks, STD2 N3 SM-Bank 38 MC

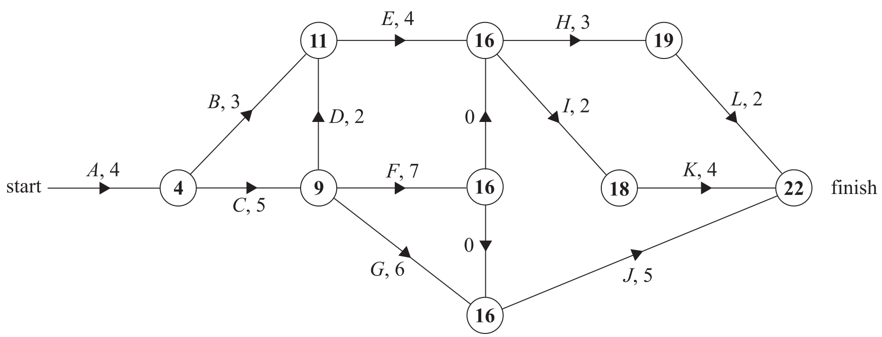
Critical Path is the longest

$\therefore DEGIK$

$\Rightarrow D$

3. Networks, STD2 N3 2011 FUR1 8 MC

Scanning forward:



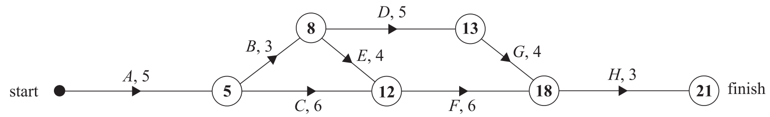
$\therefore$ The critical path is $ACFIK$.

$\Rightarrow D$

4. Networks, STD2 N3 2015 FUR1 9 MC

Consider option B,

Scanning forwards:



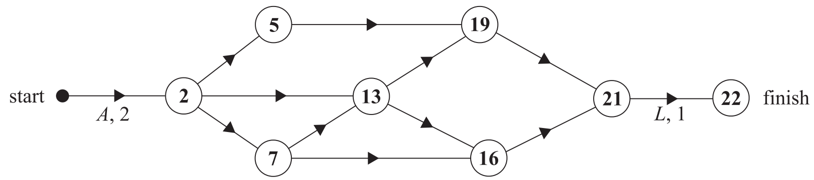
This network matches the table information.

$\Rightarrow B$

5. Networks, STD2 N3 2008 FUR1 8-9 MC

Part 1

Scanning forwards:



EST for Activity K

$= $ Duration $ACFI$

$= 2+5+6+3$

$= 16$

$\Rightarrow C$

Part 2

Original critical path: $ACFHJL$ (22 days)

♦♦ Mean mark of Part 2 was 35%.

Consider option A,

New critical path: $ABDJL$ (22 days)

$\Rightarrow A$

6. Networks, STD2 N3 2011 FUR1 7 MC

Maximum speaking time on phone

$= 40 -$ duration of M
$= 40 - 16$
$= 24$ minutes
$\Rightarrow\ D$

♦♦ *Mean mark 33%.*
MARKER'S COMMENT: Many students incorrectly answered the earliest starting time, C.

7. Networks, STD2 N3 2014 FUR1 8 MC

If any activity on a critical path takes longer, the project completion time increases by the equivalent amount of time (although the reverse is not true).

♦ *Mean mark 40%.*

$\Rightarrow\ D$

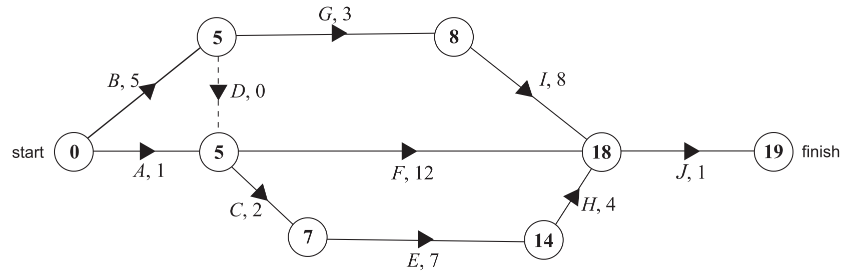
8. Networks, STD2 N3 2018 FUR1 5 MC

Activities not on critical path can be delayed.
Scanning forward: two critical paths exist (15 weeks)
$ADHK$ and $BFJK$
$\therefore$ Activity C, E, G and I could be delayed
$\Rightarrow B$

♦ *Mean mark 43%.*

9. Networks, STD2 N3 2006 FUR1 9 MC

Scanning forward:



Critical path:
$\Rightarrow BDCEHJ$ (19 hours)

Other routes not through B,
$ACEHJ$ (15 hours), AFJ (14 hours)

$\therefore$ Activity B could be reduced by 4 hours without a new critical path emerging.

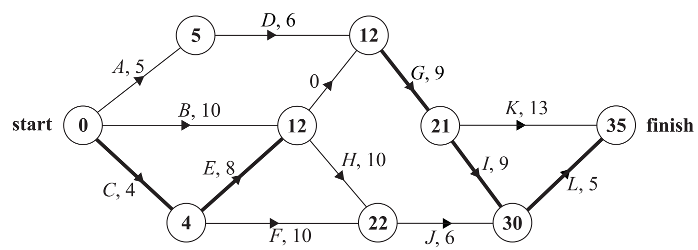
$\Rightarrow C$

♦♦♦ *Mean mark 17%.*
MARKER'S COMMENT: When choosing an activity to crash, take care that a new critical path is not created.

10. Networks, STD2 N3 2010 FUR1 8 MC

Scanning forward:

♦ *Mean mark 41%.*



Critical path is $CEGIL$
$\Rightarrow\ D$

11. Networks, STD2 N3 2012 FUR1 8 MC

Two critical paths AFH and $BCFH \Rightarrow$ 18 hours
Paths AEG and $BCEG \Rightarrow$ 17 hours
Reducing A and B by 1 hour each reduces each path above by 1 hour.

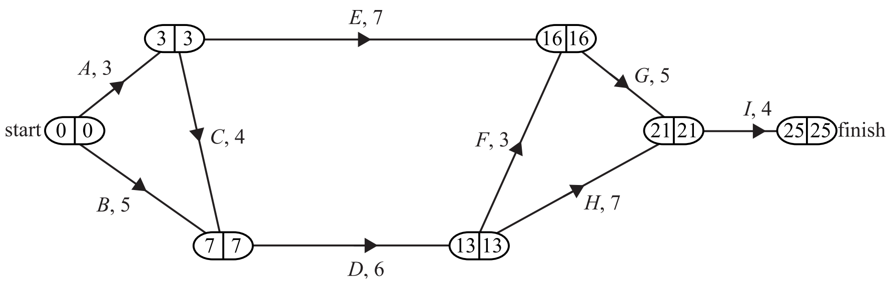
♦♦ Mean mark 30%.

Need to reduce AFH and $BCFH$ by 1 hour to 16 hours.
$\Rightarrow$ Reducing either F or H by 1 hour brings the critical path down to 16 hours.
$\Rightarrow C$

12. Networks, STD2 N3 2019 FUR2-N 2

a. D, G and I

b. Scanning forwards and backwards:



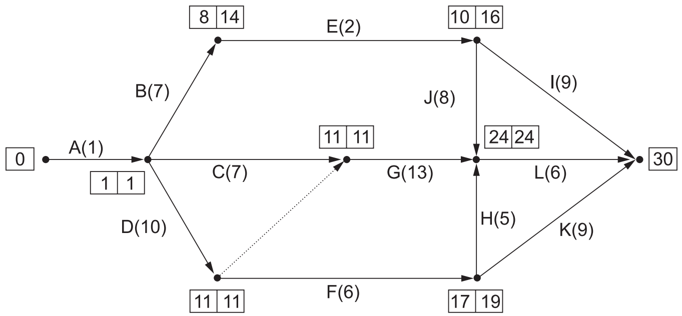
Critical Path: $ACDFGI$

c. LST (activity B) $= 7 - 5$
$= 2$ weeks

13. Networks, STD2 N3 SM-Bank 31

a. Activity C and D.

b.



Critical path $A - D - G - L$

$= 1 + 10 + 13 + 6$

$= 30$ days

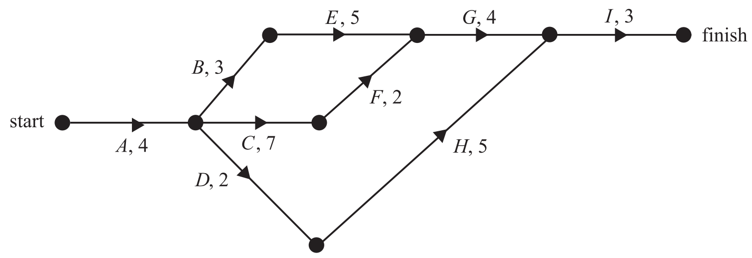
c. Float time of Activity E

$= 14 - 8$

$= 6$ days

14. Networks, STD2 N3 SM-Bank 33

i. Sketch the network:



ii. Completion time of possible paths:

$ABEGI = 4 + 3 + 5 + 4 + 3 = 19$ hours

$ACFGI = 4 + 7 + 2 + 4 + 3 = 20$ hours

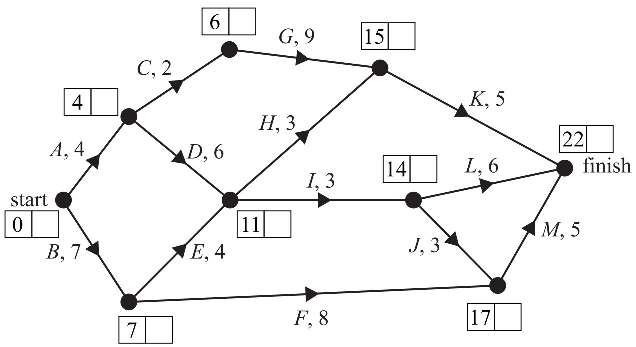
$ADHI = 4 + 2 + 5 + 3 = 14$ hours

$\therefore$ Critical path is $ACFGI$.

$\therefore$ Minimum completion time $= 20$ hours

15. Networks, STD2 N3 FUR1 2020 6

Scanning forward:



Critical path $= BEIJM$

$= 7 + 4 + 3 + 3 + 5$

$= 22$ days

16. Networks, STD2 N3 2012 FUR2 2

a. EST of $E = 10 + 2$

$= 12$ days

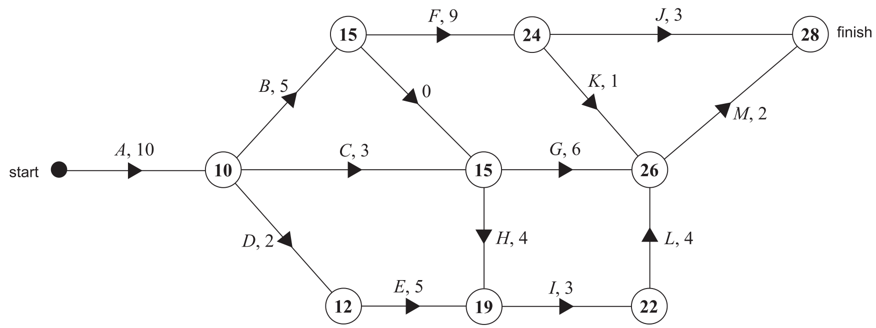
b. F has B as a predecessor while G and H have B and C as predecessors.

Since there cannot be 2 activities called B, a zero duration activity is drawn as an extension of B to show that it is also a predecessor of G and H.

♦ *Mean mark of all parts (combined) 47%.*

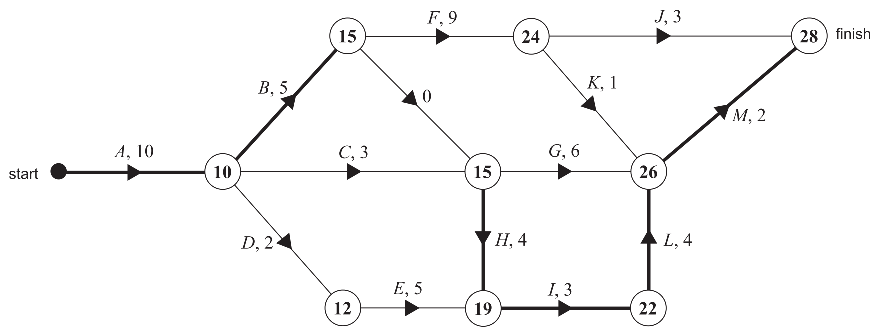
c. Scanning forwards:

♦♦ *"Few students" were able to correctly deal with the zero duration activity in part (c).*



EST for $H = 15$ days

d. The critical path is $ABHILM$



e. The shortest time to complete all the activities

$= 10 + 5 + 4 + 3 \ + 4 + 2$
$= 28$ days

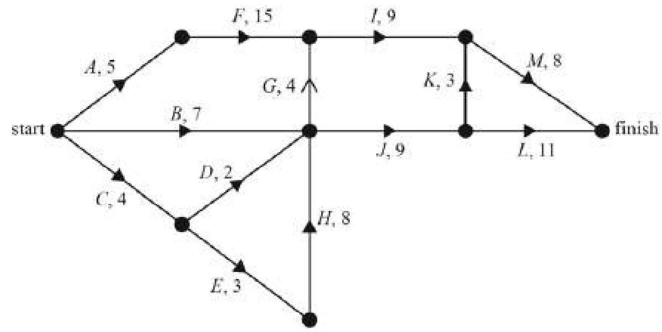
MARKER'S COMMENT: A correct calculation based on an incorrect critical path in part (d) gained a *consequential mark* here. Show your working!.

$\therefore$ LST of $J = 28 - 3$
$= 25$ days

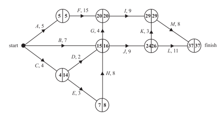
---

17. Networks, STD2 N3 2013 FUR2 2

a.



b. Scanning forwards and backwards:



EST for Activity H
$= 4 + 3$
$= 7$ hours

c.i. $AFIM$

♦♦ *Mean mark of parts (c)-(e) (combined) was 40%.*

c.ii. LST of $G = 20 - 4 = 16$ hours
LST of $D = 16 - 2 = 14$ hours

d. The statement will only be true if the time reduced activity is on the critical path $AFIM$.

MARKER'S COMMENT: Most students struggled with part (d).

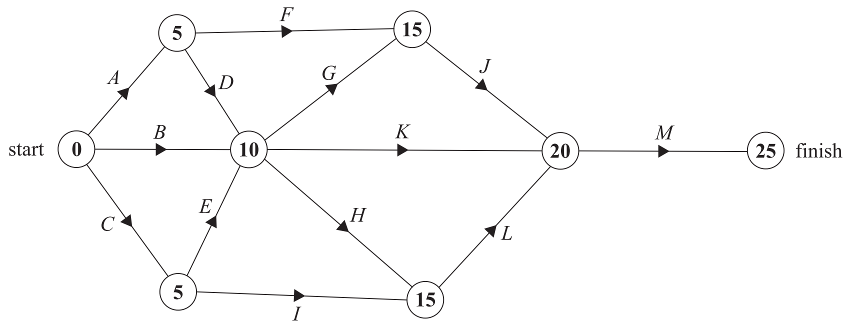
e. $AFIM$ is 37 hours.
If F is reduced by 2 hours, the new critical path is $CEHGIM$ (36 hours)
$\therefore$ Minimum completion time $= 36$ hours

18. Networks, STD2 N3 SM-Bank 48

i. A, B, C, D, E, H and I must be completed before L.

$\therefore$ 7 activities need to be completed.

ii. Scanning forward (all activities take 5 hours):



Activity F and K are not on any critical path.

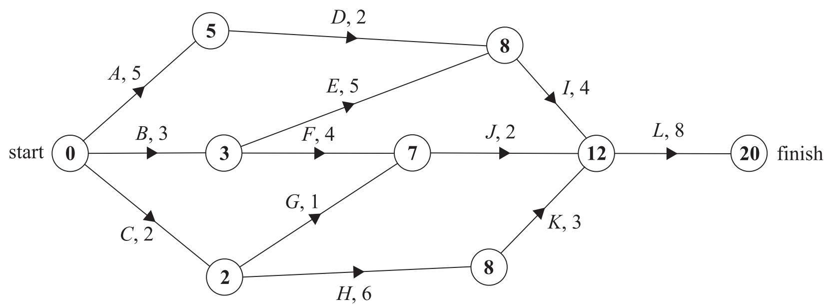
$\therefore$ Reducing either will not change the completion time for the project.

19. Networks, STD2 N3 SM-Bank 47

i. I and J

$\therefore$ 2 activities have exactly two immediate predecessors.

ii. Scanning forwards:



$\therefore$ Initial critical path $BEIL$

iii. If E reduced by 1 hour, critical path $BEIL$ reduces to 19 hours.

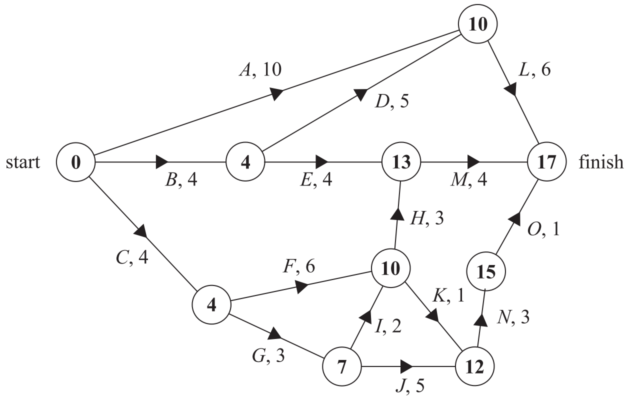
$\therefore$ Other critical paths of 19 hours are:

$ADIL = 5 + 2 + 4 + 8 = 19$

$CHKL = 2 + 6 + 3 + 8 = 19$

20. Networks, STD2 N3 SM-Bank 49

i. Scanning forward:



$\therefore$ EST for $N = CGJ$

$= 4 + 3 + 5$

$= 12$

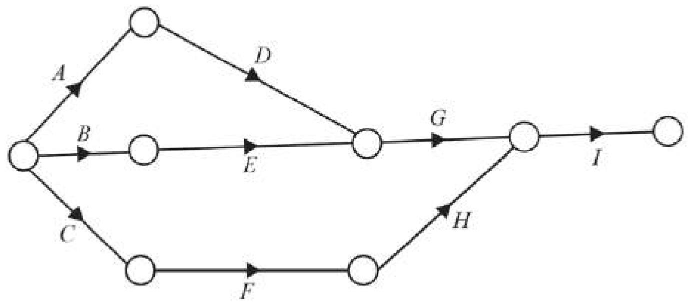
ii. Critical path is: $CFHM$

$\therefore$ 4 activities can't be delayed.

---

21. Networks, STD2 N3 2006 FUR2 3

a.



b. Since no time information, possible critical paths are:

$ADGI$, $BEGI$ or $CFHI$ (all have 4 activities)

$\therefore$ Non-critical activities

$= 9 - 4 = 5$

c. Critical activities have zero float time.

$\Rightarrow$ A and C are non-critical.

$\therefore$ $BEGI$ is the critical path.

♦ *Mean mark of parts (c)-(e) (combined) was 36%.*

d. Duration of $I = 19 - 12$

$= 7$ hours

e. Maximum time for F and H

$=$ LST of I − duration C − slack time of C

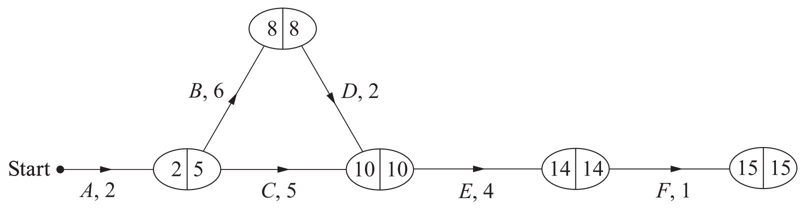
$= 12 - 3 - 1$

$= 8$ hours

♦♦ **MARKER'S COMMENT:** Many students incorrectly answered 9 hours in part (e).

22. Networks, STD2 N3 2019 HSC 26

a.



Scanning forwards:

Minimum time $= 2+6+2+4+1 = 15$ hours

(Scanning forwards and backwards is highly recommended but not required in the network diagram.)

b. Critical Path is $ABDEF$.

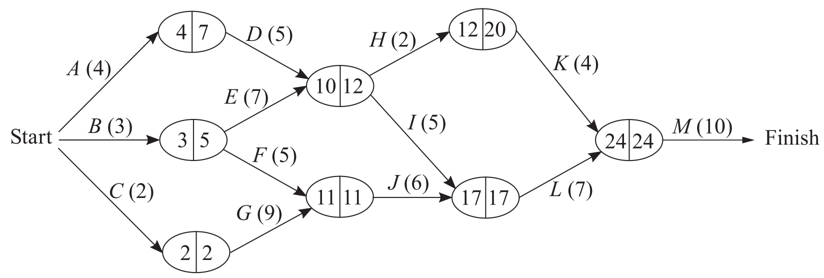
Non-critical activity is C.

♦♦ *Mean mark 30%.*

$$\begin{aligned}\text{Float time} &= 5-2\\ &= 3 \text{ hours}\end{aligned}$$

23. Networks, STD2 N3 SM-Bank 39

i. Scanning forwards and backwards



EST (activity J) $= 11$ hours

ii. $$\begin{aligned}\text{LST (activity } H) &= \text{LFT } (H) - \text{weight } (H)\\ &= 20-2\\ &= 18 \text{ hours}\end{aligned}$$

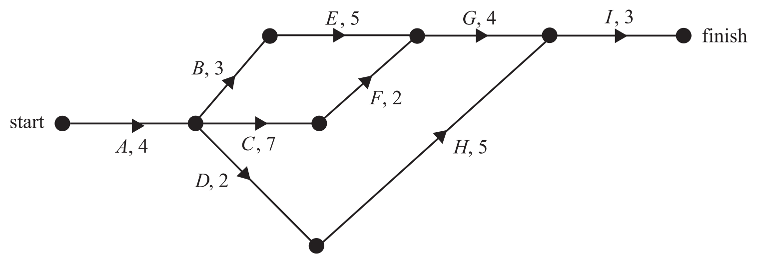
iii. $$\begin{aligned}\text{Float time } (I) &= \text{LST } (I) - \text{EST } (I)\\ &= 12-10\\ &= 2 \text{ hours}\end{aligned}$$

iv. Critical Path is $CGJLM$

$$\begin{aligned}\therefore \text{ Minimum time } &= 2+9+6+7+10\\ &= 34 \text{ hours}\end{aligned}$$

24. Networks, STD2 N3 SM-Bank 49

i. Sketch the network:



ii. Completion time of paths:

$ABEGI = 4+3+5+4+3 = 19$ hours

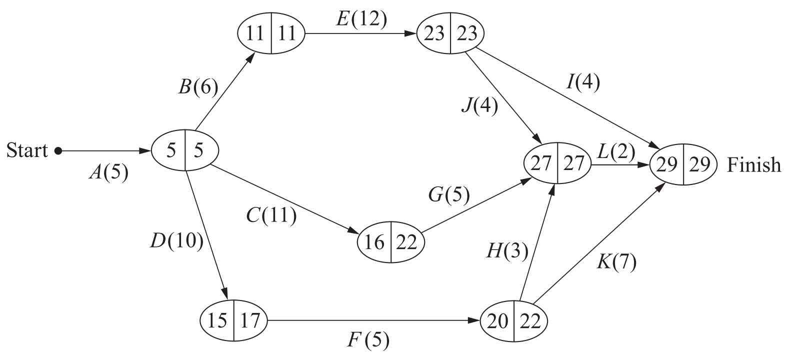
$ACFGI = 4+7+2+4+3 = 20$ hours

$ADHI = 4+2+5+3 = 14$ hours

$\therefore$ Minimum completion time (critical path) $= 20$ hours

25. Networks, STD2 N3 SM-Bank 29

Scanning forwards and backwards:



Critical Path is $ABEJL$ (29 days)

$\Rightarrow$ Critical path is 2 days less than 31 day deadline

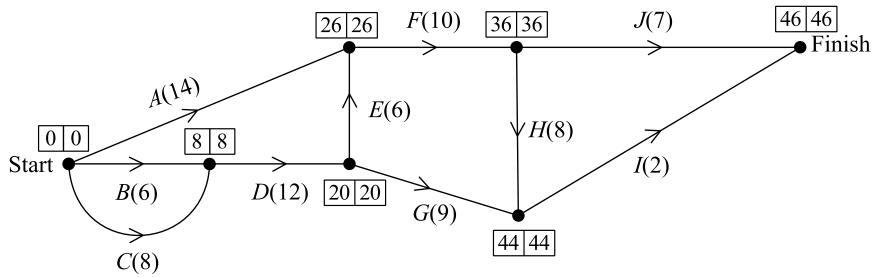
$\therefore$ Float time of Activity $G = (22 - 16) + 2$

$= 8$ days

26. Networks, STD2 N3 2020 HSC 26

a. Scan forwards and backwards (to answer all parts):

♦♦ *Mean mark part (a) 31%.*



Minimum time $= 46$ minutes

b. $C - D - E - F - H - I$

Mean mark part (b) 51%.

c. Scan backwards (see above):

♦♦ *Mean mark part (c) 22%.*

Activity	*Earliest start time* (minutes)	*Float time* (minutes)
A	0	Float = 12 - 0 = 12
G	20	Float = 35 - 20 = 15

27. Networks, STD2 N3 2021 FUR1 6

Consider all possible network paths:

$ADHJ \ \rightarrow\ $ 18 hours

$BEJ \ \rightarrow\ (12+x)$ hours

$CGFEJ \ \rightarrow\ (15+x)$ hours

$CGIJ \ \rightarrow\ $ 18 hours

Minimum completion time $=18$ hours

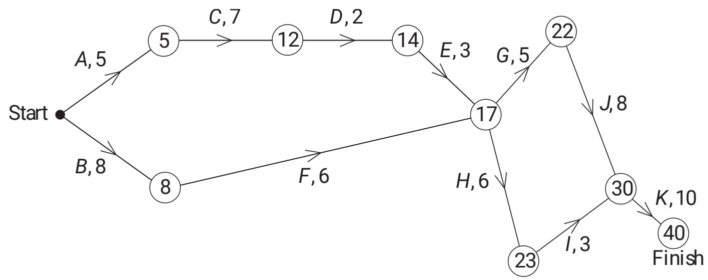
$\Rightarrow$ No path can be longer than 18 hours

$\therefore\ x_{\text{max}}=3$ hours

28. Networks, STD2 N3 2021 HSC 36

a. Scanning forwards:

♦ *Mean mark part (a) 37%.*



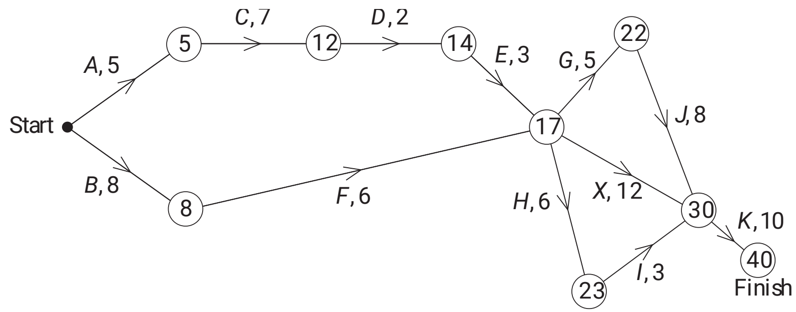
Minimum completion time $=40$ minutes

♦ *Mean mark part (b) 44%.*

b. Critical Path: $ACDEGJK$

♦♦ *Mean mark part (c) 32%.*

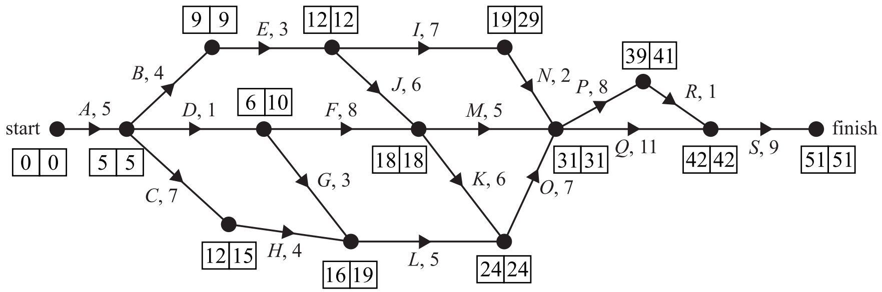
c.



$$\begin{aligned}\text{Float time }(X)&=18-17\\&=1\text{ minute}\end{aligned}$$

29. Networks, STD2 N3 SM-Bank 41

i. Scanning forwards then backwards:

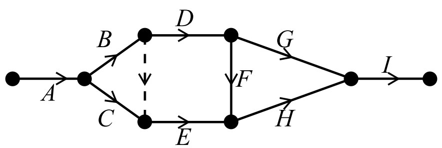


Critical Path: $A-B-E-J-K-O-Q-S$

ii. Using the scanned diagram, activity G and N have a float time of 10 hours.

30. Networks, STD2 N3 SM-Bank 44

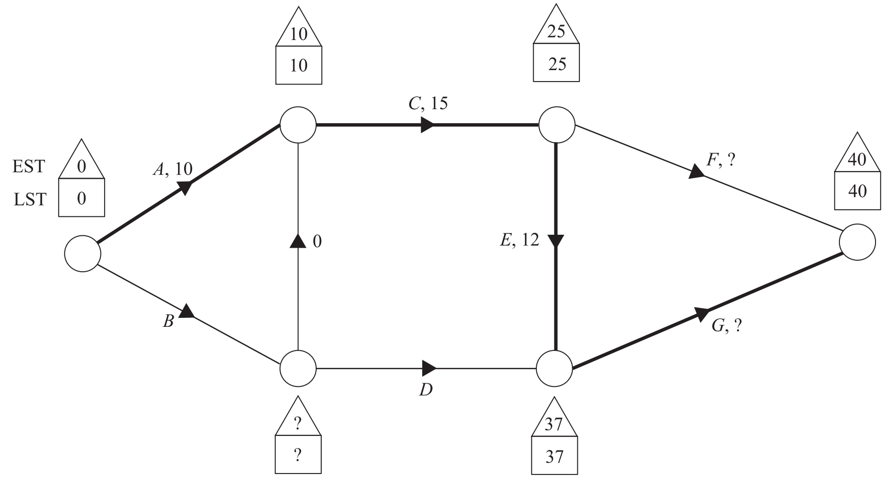
Sketch network diagram:



The dummy activity needs to be drawn from the end of activity B to the start of activity E.

31. Networks, STD2 N3 EQ-Bank 18

Sketch network:



Critical path added to network (where EST = LST)

G is on critical path $\Rightarrow$ no float

$\therefore$ Duration of $G = 40 - 37 = 3$ weeks

$$\begin{aligned}\text{Duration of } F &= 40 - \text{EST of } F - \text{float}\\ &= 40 - 25 - 6\\ &= 9 \text{ weeks}\end{aligned}$$

Float of $D = 9$ weeks (given)

$$\begin{aligned}\therefore \text{ Duration of } D &= \text{LST of } G - \text{EST of } D - 9\\ &= 37 - \text{EST of } D - 9\\ &= 28 - \text{EST of } D\end{aligned}$$

EST of $D =$ Duration of B (B has no prerequisites)

Duration of $B +$ Duration of $D = 28$ weeks

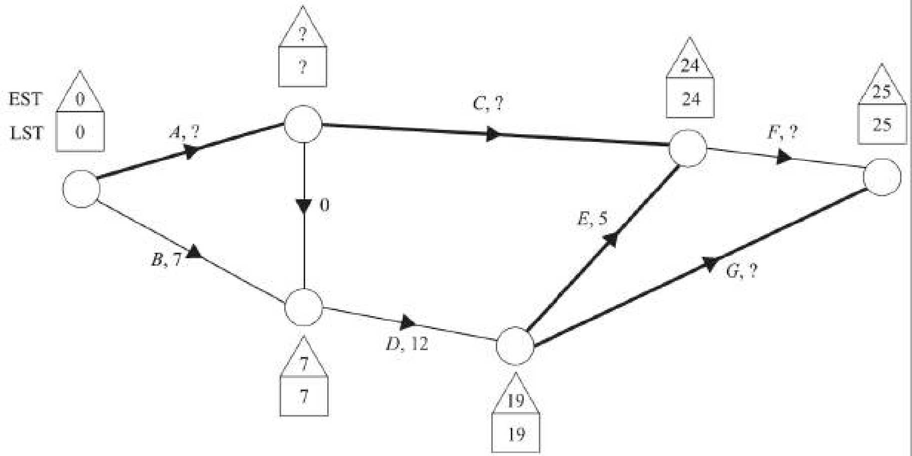
$\therefore$ Possible durations of B and D are:

$B = 1$ week, $D = 27$ weeks

*Understand why there are many possibilities for the duration of B and D, provided they add up to 28 weeks and B is not longer than 10 weeks (or a new critical path is created).

32. Networks, STD2 N3 EQ-Bank 19

Sketch the network:



Critical path information included in the network (where EST $=$ LST)

F is on critical path $\Rightarrow$ no float

$\therefore$ Duration of $F = 25 - 24 = 1$ day

Duration of $G =$ LST of next activity $-$ EST of G $-$ float

$= 25 - 19 - 2$

$= 4$ days

The float of C is 4 days (given)

$\therefore$ Duration of $C =$ LST of $F -$ EST of $C - 4$

$= 24 -$ EST of $C - 4$

$= 20 -$ EST of C

EST of $C =$ Duration of A (A has no prerequisites)

$\Rightarrow$ Duration of $A +$ Duration of $C = 20$ days

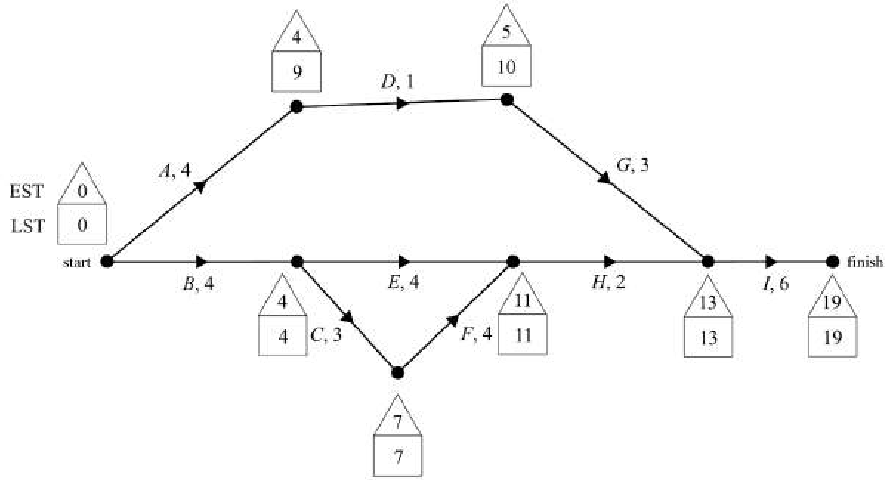
$\therefore$ Possible durations of A and C are:

$A = 5$ days, $C = 15$ days

*Understand why there are many possibilities for the duration of A and C, provided they add up to 20 days and A is not longer than 7 days (or a new critical path is created).

33. Networks, STD2 N3 2007 FUR2 4

a. Scanning forwards and backwards:



$BCFHI$ is the critical path.

$\therefore$ Minimum time $= 4 + 3 + 4 + 2 + 6$

$= 19$ weeks

♦ Mean mark of all parts (combined) 40%.

b. EST of $D = 4$

LST of $D = 9$

$\therefore$ Float time of $D = 9 - 4$

$= 5$ weeks

c. A, E, and G are not currently on the critical path, therefore reducing their time will not result in an earlier completion time.

d. Reduce C and F by 2 weeks each.

However, a new critical path created: $BEHI$ (16 weeks)

$\therefore$ Also reduce E by 1 week.

$\therefore$ Minimum completion time $= 15$ weeks

e. Additional cost $= 5 \times \$5000$

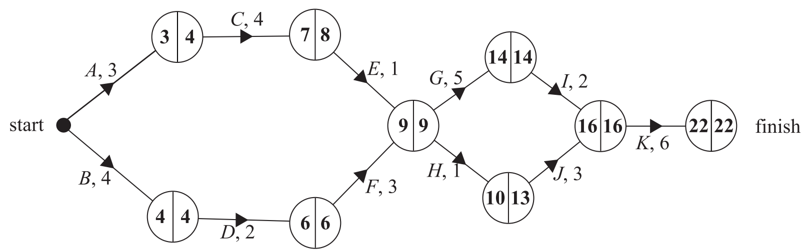
$= \$25\,000$

34. Networks, STD2 N3 2009 FUR2 4

a. 7 weeks

♦ *Mean mark of all parts (combined): 44%.*

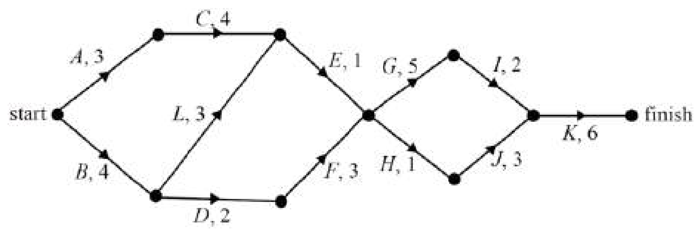
b. Scanning forwards and backwards



Critical Path is $BDFGIK$

c. H or J can be delayed for a maximum of 3 weeks.

d.



e. The new critical path is $BLEGIK$.

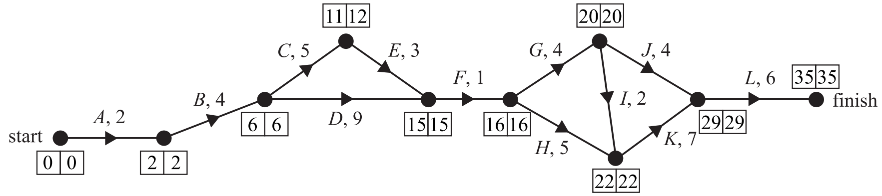
$\Rightarrow$ Activity L now takes 7 weeks.

$\therefore$ Time for completion

$$= 4 + 7 + 1 + 5 + 2 + 6$$
$$= 25 \text{ weeks}$$

35. Networks, STD2 N3 2019 FUR2 3

a. Scanning forwards and backwards:



Critical path: $ABDFGIKL$

$\therefore$ 8 activities

b. LST for activity $E = 12$ weeks (i.e. start of 13th week)

c. Consider float times of all activities not on critical path.

$J - 5, H - 1, E - 1, C - 1$

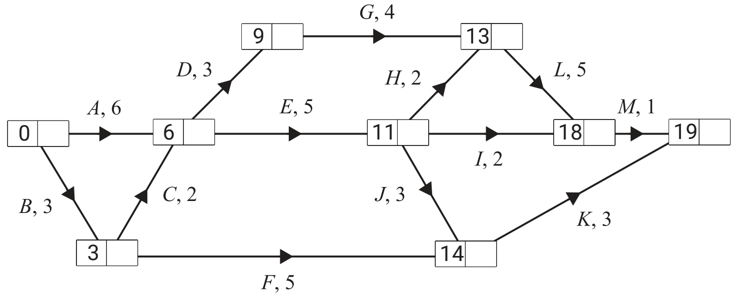
$\therefore$ Activity J has the largest float time.

d. Critical path after reducing $CDGHK$ by 2 weeks is $ABDFGIKL$.

$$\therefore \text{ Minimum time } = 2 + 4 + 7 + 1 + 2 + 2 + 5 + 6$$
$$= 29 \text{ weeks}$$

36. Networks, STD2 S3 SM-Bank 50

a. Scan forwards:



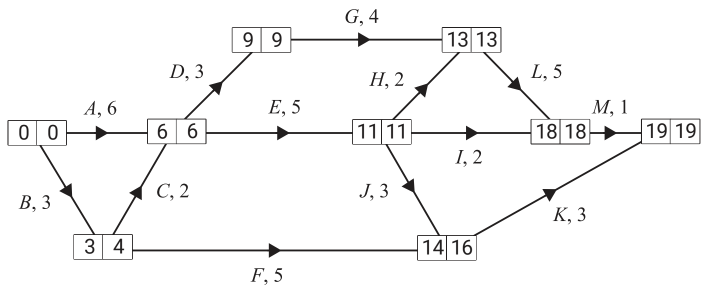
EST (activity K)

$= A\ E\ J$

$= 6 + 5 + 3$

$= 14$ weeks

b. Scan backwards:



Critical paths: $ADGLM$ and $AEHLM$

$\therefore$ 7 activities have no float time: $ADEGHLM$

c. There are 9 possible paths

$ADGLM$ (19), $AEHLM$ (19), $AEIM$ (14)

$AEJK$ (17), $BCEIM$ (13), $BCEJK$ (16)

$BCEHLM$ (18), $BCDGLM$ (18), BFK(11)

Consider the 5 paths with completions over 16 weeks

$\rightarrow$ all contain A or L or both.

Consider $ADGLM$, $AEHLM$ (contains both A and L)

$\rightarrow$ reduce $L \times 2$, $A \times 1$ to reach 16 weeks

$\rightarrow$ cheaper than $L \times 1$, $A \times 2$

Consider $BCEHLM$, $BCDGLM$ (both contain L only)

$\rightarrow$ reduce $L \times 2$ to reach 16 weeks

Consider $AEJK$ (contains A only)

$\rightarrow$ reduce $A \times 1$ reach 16 weeks.

$\therefore$ Minimum cost to reduce time to 16 weeks

$= 2 \times 120\ 000 + 1 \times\ 140\ 000$

$= \$380\ 000$